

Support of the Fake-Degree Schur Sum at a Root of Unity

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Abstract

For $n \geq 1$ and $e \geq 2$, define

$$q_e := \sum_{\lambda \vdash n} \tilde{f}_\lambda(\zeta_e) s_\lambda \in \Lambda_n \otimes \mathbb{Z}[\zeta_e],$$

where $\tilde{f}_\lambda(t)$ is the fake degree of λ and ζ_e is a primitive e -th root of unity. We prove that q_e is supported on power-sum monomials p_μ with

$$\mu = (e^{\lfloor n/e \rfloor}, \nu), \quad \nu \vdash n \bmod e.$$

The proof is a two-line consequence of the Chevalley–Molien formula for the graded character of the S_n -coinvariant algebra. As a corollary, q_e factors as $p_e^{\lfloor n/e \rfloor} \cdot F_{n \bmod e}$ for a computable $F_{n \bmod e} \in \Lambda_{n \bmod e} \otimes \mathbb{Z}[\zeta_e]$.

1 Statement

Let Λ_n denote the space of homogeneous symmetric functions of degree n over $\mathbb{Q}$. For $\lambda \vdash n$, the *fake degree* of λ is

$$\tilde{f}_\lambda(t) := \sum_{T \in \text{SYT}(\lambda)} t^{\text{maj}(T)} = t^{n(\lambda)} \frac{[n]_t!}{\prod_{c \in \lambda} [h(c)]_t},$$

where $n(\lambda) = \sum_i (i-1)\lambda_i$ and $h(c)$ is the hook length of the cell c . Equivalently, $\tilde{f}_\lambda(t)$ is the graded multiplicity of the irreducible V_λ in the coinvariant algebra of S_n .

Fix $e \geq 2$ and let ζ_e be a primitive e -th root of unity. Set

$$q_e := \sum_{\lambda \vdash n} \tilde{f}_\lambda(\zeta_e) s_\lambda \in \Lambda_n \otimes \mathbb{Z}[\zeta_e].$$

Theorem 1. *Write $q_e = \sum_\mu c_\mu p_\mu$ in the power-sum basis. Then $c_\mu \neq 0$ if and only if μ has the form*

$$\mu = (e^{\lfloor n/e \rfloor}, \nu), \quad \nu \vdash n \bmod e.$$

Equivalently, writing $\langle \cdot, \cdot \rangle$ for the Hall inner product, $\langle q_e, p_\mu \rangle = 0$ unless μ has $\lfloor n/e \rfloor$ parts equal to e and the remaining parts sum to $n \bmod e$.

2 Chevalley–Molien Preliminaries

Let $V = \mathbb{C}x_1 + \cdots + \mathbb{C}x_n$ carry the natural S_n -action by permutation of coordinates, and let $S = \mathbb{C}[V^*] = \mathbb{C}[x_1, \dots, x_n]$ with the induced graded action. The graded character

$$\mathrm{Tr}(g \mid S)(t) := \sum_{d \geq 0} \mathrm{Tr}(g \mid S_d) t^d$$

is computed by Molien’s formula:

$$\mathrm{Tr}(g \mid S)(t) = \frac{1}{\det_V(1 - tg)}.$$

For $g \in S_n$ of cycle type $\mu = (\mu_1, \dots, \mu_\ell)$, the permutation matrix g has, on each cycle of length μ_i , eigenvalues $\{\zeta_{\mu_i}^0, \dots, \zeta_{\mu_i}^{\mu_i-1}\}$. Since $\prod_{k=0}^{\mu_i-1} (1 - t\zeta_{\mu_i}^k) = 1 - t^{\mu_i}$, we obtain

$$\mathrm{Tr}(g \mid S)(t) = \frac{1}{\prod_{j \in \mu} (1 - t^j)}. \quad (1)$$

Chevalley’s theorem gives $S \cong S^{S_n} \otimes_{\mathbb{C}} H$ as graded S_n -modules, where $H := S/(S_+^{S_n})$ is the coinvariant algebra and $S^{S_n} = \mathbb{C}[e_1, \dots, e_n]$ (with $\deg e_i = i$) acts trivially. Consequently

$$\mathrm{Tr}(g \mid S)(t) = \mathrm{Tr}(g \mid S^{S_n})(t) \cdot \mathrm{Tr}(g \mid H)(t) = \frac{1}{\prod_{i=1}^n (1 - t^i)} \cdot \mathrm{Tr}(g \mid H)(t),$$

where we used that g acts trivially on S^{S_n} so $\mathrm{Tr}(g \mid S^{S_n})(t) = \mathrm{Hilb}(S^{S_n}; t) = \prod_{i=1}^n (1 - t^i)^{-1}$. Solving for $\mathrm{Tr}(g \mid H)$ and combining with (1) gives

$$\mathrm{Tr}(g \mid H)(t) = \frac{\prod_{i=1}^n (1 - t^i)}{\prod_{j \in \mu} (1 - t^j)}. \quad (2)$$

On the other hand, H decomposes as $\bigoplus_{\lambda \vdash n} V_\lambda \otimes W_\lambda(t)$ with $W_\lambda(t)$ a graded multiplicity space of graded dimension $\tilde{f}_\lambda(t)$, so

$$\mathrm{Tr}(g \mid H)(t) = \sum_{\lambda \vdash n} \tilde{f}_\lambda(t) \chi^\lambda(\mu). \quad (3)$$

Combining (2) and (3) yields the identity we need.

Lemma 2 (Chevalley–Molien for fake degrees). *For every $n \geq 1$ and $\mu \vdash n$,*

$$\Psi_\mu(t) := \sum_{\lambda \vdash n} \tilde{f}_\lambda(t) \chi^\lambda(\mu) = \frac{\prod_{i=1}^n (1 - t^i)}{\prod_{j \in \mu} (1 - t^j)}$$

as elements of $\mathbb{Z}[t]$.

Proof. Immediate from (2) and (3). The ratio on the right is a polynomial (with integer coefficients) because it is the graded trace on the finite-dimensional coinvariant algebra H . $\square$

3 Proof of Theorem 1

For $\lambda \vdash n$, the Schur expansion in the power-sum basis is

$$s_\lambda = \sum_{\mu \vdash n} z_\mu^{-1} \chi^\lambda(\mu) p_\mu, \quad z_\mu := \prod_i i^{m_i(\mu)} m_i(\mu)!,$$

where $m_i(\mu)$ is the multiplicity of i in μ . Substituting into the definition of q_e and swapping sums,

$$q_e = \sum_{\mu \vdash n} z_\mu^{-1} \left(\underbrace{\sum_{\lambda \vdash n} \tilde{f}_\lambda(\zeta_e) \chi^\lambda(\mu)}_{= \Psi_\mu(\zeta_e) \text{ by Lemma 2}} \right) p_\mu.$$

Hence

$$c_\mu = z_\mu^{-1} \Psi_\mu(\zeta_e) = \frac{1}{z_\mu} \cdot \frac{\prod_{i=1}^n (1 - \zeta_e^i)}{\prod_{j \in \mu} (1 - \zeta_e^j)}. \quad (4)$$

The factor $(1 - \zeta_e^m)$ vanishes if and only if $e \mid m$. Since $\Psi_\mu(t)$ is a polynomial (Lemma 2), we may factor $\Psi_\mu(t) = \Phi_e(t)^{v_e(\mu)} Q_\mu(t)$ with $Q_\mu(\zeta_e) \neq 0$, where

$$v_e(\mu) = |\{i \in [1, n] : e \mid i\}| - |\{j \in \mu : e \mid j\}| = \lfloor n/e \rfloor - k_e(\mu), \quad (5)$$

with $k_e(\mu) := |\{j \in \mu : e \mid j\}|$.

Lemma 3. $c_\mu = 0$ if and only if $k_e(\mu) < \lfloor n/e \rfloor$.

Proof. From (4), $c_\mu = 0$ iff $\Psi_\mu(\zeta_e) = 0$ iff $v_e(\mu) > 0$, iff $k_e(\mu) < \lfloor n/e \rfloor$. We need $v_e(\mu) \geq 0$: parts of μ divisible by e are each at least e , so their number $k_e(\mu)$ satisfies $e \cdot k_e(\mu) \leq n$, hence $k_e(\mu) \leq \lfloor n/e \rfloor$. $\square$

Proof of Theorem 1. By Lemma 3, $c_\mu \neq 0$ iff $k_e(\mu) = \lfloor n/e \rfloor$. Let $k := \lfloor n/e \rfloor$. Suppose $k_e(\mu) = k$; write the parts of μ divisible by e as $ea_1, \dots, ea_k$ with $a_i \geq 1$. Their sum is $e \sum_i a_i$, and the remaining parts sum to $n - e \sum_i a_i \geq 0$, so $\sum_i a_i \leq n/e$, i.e., $\sum_i a_i \leq k$. Combined with $\sum_i a_i \geq k$, we get $\sum_i a_i = k$, so every $a_i = 1$.

Thus the parts of μ divisible by e are exactly $\lfloor n/e \rfloor$ copies of e . Let ν be the partition formed by the remaining parts. Then $|\nu| = n - ek = n \bmod e < e$, so all parts of ν are strictly less than e (in particular not multiples of e). Hence $\mu = (e^{\lfloor n/e \rfloor}, \nu)$ with $\nu \vdash n \bmod e$.

Conversely, if $\mu = (e^{\lfloor n/e \rfloor}, \nu)$ with $\nu \vdash n \bmod e$, then $k_e(\mu) = \lfloor n/e \rfloor$ (the parts of ν are $\leq n \bmod e < e$, hence not multiples of e), so $v_e(\mu) = 0$ and $c_\mu \neq 0$. $\square$

4 Corollary: Product Structure and Explicit Formula

Since $p_{(e^k, \nu)} = p_e^k \cdot p_\nu$ in Λ , Theorem 1 immediately gives:

Corollary 4. With $k := \lfloor n/e \rfloor$ and $\rho := n \bmod e$,

$$q_e = p_e^k \cdot F_e,$$

where $F_e \in \Lambda_\rho \otimes \mathbb{Z}[\zeta_e]$ has power-sum expansion

$$F_e = \sum_{\nu \vdash \rho} c_{(e^k, \nu)} p_\nu.$$

We can compute the coefficients $c_{(e^k, \nu)}$ in closed form. For $\mu = (e^k, \nu)$,

$$z_\mu = e^k \cdot k! \cdot z_\nu, \quad \Psi_\mu(\zeta_e) = \frac{\prod_{i=1}^n (1 - \zeta_e^i)}{(1 - \zeta_e^e)^k \prod_{j \in \nu} (1 - \zeta_e^j)}.$$

Both numerator and denominator vanish to order k at ζ_e ; factoring out $\Phi_e(t)^k$ and evaluating,

$$\lim_{t \rightarrow \zeta_e} \frac{\prod_{m=1}^k (1 - t^{em})}{(1 - t^e)^k} = \prod_{m=1}^k \left(\lim_{t \rightarrow \zeta_e} \frac{1 - t^{em}}{1 - t^e} \right) = \prod_{m=1}^k m = k!,$$

using $\lim_{u \rightarrow 1} (1 - u^m)/(1 - u) = m$ with $u = t^e$. Writing $\alpha_r := 1 - \zeta_e^r$ for $r \in \{1, \dots, e-1\}$ and setting $b_r := k + \mathbf{1}[r \leq \rho]$, the surviving factors give

$$c_{(e^k, \nu)} = \frac{1}{e^k z_\nu} \prod_{r=1}^{e-1} \alpha_r^{b_r - m_r(\nu)}. \quad (6)$$

Remark 5. Formula (6) matches the values computed in Section 5:

- At $(n, e) = (5, 3)$: $k = 1$, $\rho = 2$, so $b_1 = b_2 = 2$. For $\nu = (2)$: $m_1 = 0, m_2 = 1$, so $c_{(3,2)} = \frac{1}{3 \cdot 2} \alpha_1^2 \alpha_2 = \frac{\alpha_1^2 \alpha_2}{6}$. For $\nu = (1, 1)$: $m_1 = 2, m_2 = 0$, so $c_{(3,1,1)} = \frac{1}{3 \cdot 2} \alpha_2^2 = \frac{\alpha_2^2}{6}$.
- At $(n, e) = (20, 9)$: $k = 2$, $\rho = 2$, and $\Psi_{(9,2)}(\zeta_9)$ evaluates to $162 - 162\zeta_9$, matching the $2! \cdot 81 = 162$ prefactor.

5 Computational Verification

Two independent scripts verify Theorem 1.

(a) Chevalley–Molien polynomial identity. For every $n \leq 7$ and every $\mu \vdash n$, we verified $\sum_\lambda \tilde{f}_\lambda(t) \chi^\lambda(\mu) = \prod (1 - t^i) / \prod (1 - t^j)$ as polynomials in $\mathbb{Z}[t]$. This checks Lemma 2 at $\sum_{n \leq 7} p(n) = 30$ pairs (n, μ) .

(b) Support of q_e . For each of the pairs

$$(n, e) \in \{(5, 3), (6, 3), (7, 5), (8, 3), (9, 3), (10, 4), (11, 3), (15, 4)\},$$

we computed the coefficient c_μ of p_μ in q_e over all $\mu \vdash n$ (total $\sum p(n) = 7 + 11 + 15 + 22 + 30 + 42 + 56 + 176 = 359$ tests), reducing modulo $\Phi_e(t)$. Every μ not of the shape $(e^{\lfloor n/e \rfloor}, \nu)$ gave $c_\mu = 0$, and every μ of that shape gave $c_\mu \neq 0$. Sample:

(n, e)	μ	$\Psi_\mu(\zeta_e)$ (writing $t := \zeta_e$)
(5, 3)	(3, 2)	$3 - 3t$
(5, 3)	(3, 1, 1)	$3 + 3t$
(7, 5)	(5, 2)	$5 - 5t$
(7, 5)	(5, 1, 1)	$5 + 5t$
(8, 3)	(3, 3, 2)	$18 - 18t$
(8, 3)	(3, 3, 1, 1)	$18 + 18t$
(10, 4)	(4, 4, 2)	$32 - 32t$
(10, 4)	(4, 4, 1, 1)	$32 + 32t$

The prefactors $(3, 5, 18, 32)$ match $k! \prod \alpha_r^{b_r - m_r(\nu)}$ from (6), reduced modulo Φ_e .

(c) **The (20, 9) instance.** Yesterday’s probe (2026-08-05) verified over all 627 partitions of 20: the only μ with $c_\mu \neq 0$ are $(9, 9, 2)$ with $\Psi_\mu(\zeta_9) = 162 - 162\zeta_9$ and $(9, 9, 1, 1)$ with $\Psi_\mu(\zeta_9) = 162 + 162\zeta_9$. Both match (6) with $k = 2$, $2! \cdot 81 = 162$.

6 Remarks and Cross-References

Remark 6 (Springer-regular case as a special case). When e divides n and $e \in \{\text{degrees of } S_n\} = \{1, 2, \dots, n\}$ (i.e., $e \leq n$), the setup is the Kraśkiewicz–Weyman regularity case: $\lfloor n/e \rfloor = n/e$, $n \bmod e = 0$, so the only ν is empty and $q_e = c \cdot p_e^{n/e}$ for a single coefficient c . This recovers Springer’s regular-element formula.

Remark 7 (Connection to cyclic sieving). The polynomial $\Psi_\mu(t)$ is exactly the graded character of the S_n -coinvariant ring at a permutation of cycle type μ . Reiner, Stanton and White [1] use exactly the ratio $\prod(1 - t^i) / \prod(1 - t^j)$ (evaluated at roots of unity) as the sieving polynomial in a foundational example of the cyclic sieving phenomenon. Theorem 1 is a straightforward consequence in that language: $\Psi_\mu(\zeta_e)$ counts fixed points of $\langle \sigma \rangle$ (with σ a fixed element of order e in S_n) acting on the coset space $S_n/C(\mu)$, and this count is zero unless σ can be conjugated inside a maximal torus normalizer for cycle-type μ , i.e., unless μ contains $\lfloor n/e \rfloor$ copies of e .

Remark 8 (The Albion Verschiebung route is not needed). The PROVE plan (2026-08-06) suggested closing the conjecture via Albion’s Theorem 1.3 on Verschiebung acting on Schur functions [2]: apply φ_e to both sides of a putative product identity $q_e = c p_e^k F_{n \bmod e}$, using $\varphi_e s_\lambda = 0$ unless the e -core of λ is empty. This route works but is heavier: it requires identifying the e -quotient structure that makes the sum collapse, and it recovers only the same order-of-vanishing statement that falls out of the Chevalley–Molien formula in one line. The Chevalley–Molien path is more direct because it goes through the coinvariant algebra, where the Verschiebung structure is already *built in* to the isomorphism $H \cong \mathbb{C}[S_n]/(\dots)$ via the graded character formula.

Remark 9 (Application to composite- d divisibility). Theorem 1 unlocks the mechanism behind Theorem B of [3] (unconditional $\Phi_9 \mid m_\pi$ at $(r, k) = (5, 4)$): for any triple (r, k, e) where the factorisation-count obstruction fires (no partition of r can produce $\lfloor n/e \rfloor$ copies of e in $p_\lambda[h_k]$), we obtain $\Phi_e \mid m_\pi(t)$ for all $\pi \vdash r$. This anchors the composite- d paper alongside Theorems A + B.

References

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